

P2E: EVOLVING EVALUATION SYSTEMS FOR OPEN-ENDED GENERATION

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ABSTRACT

Human preferences specify desired orderings of open-ended outputs without identifying what an evaluator must observe or how it should interpret that evidence. We present P2E, Preference-Guided Evaluator Evolution, a framework for learning executable evaluation procedures under domain-specific human standards. P2E maps existing metrics, judges, and tools to quality dimensions and acquires structured feedback on semantically distinct disagreements. This evidence grounds hypotheses about missing observations, unused evidence, or inappropriate criteria. Within P2E, ERA evolves artifact representations, rubrics, evaluator modules, tools, evidence routing, and fusion while holding the base model fixed. Mechanism hypotheses guide connected changes, whereas complete programs are the units of selection. This separation permits evidence-guided continuation without equating an unsuccessful implementation with a refuted hypothesis. The research scope spans structured visual artifacts, images and 3D, text generation, and automated research. We specify matched-budget studies of feedback acquisition and evolution, followed by independent tests of alignment and selection, critic-guided refinement, and planned reward-guided training. Whether an evolved evaluator agrees better with humans and whether it improves generation remain distinct empirical questions.

1 INTRODUCTION

An appealing image may omit a requested object. A fluent summary may remove a qualification that changes the source’s meaning. A research proposal may offer an interesting hypothesis without an experiment capable of testing it. These are evaluation problems in open-ended generation: multiple outputs are reasonable, quality has several dimensions, and no single deterministic verifier defines the complete ordering. Existing metrics and model-based judges provide useful but different evidence about that ordering (Fabbri et al., 2021; Xu et al., 2023; Ghosh et al., 2023; Liu et al., 2023).

The central difficulty is that a preference error does not identify its cause. A summary evaluator might never inspect the source qualification, observe it without passing it to the judge, or judge faithfulness correctly but give fluency too much weight. The same wrong ordering is compatible with each mechanism. Adding another score does not resolve this ambiguity when sources measure different constructs or share a blind spot. Work on model-based judging and human-authored criteria motivates examining what supports a judgment (Zheng et al., 2023; Shankar et al., 2024). This leads to two linked questions: *which human judgments reveal gaps in evaluation*, and *how should an evaluator evolve in response?*

We introduce P2E, Preference-Guided Evaluator Evolution, for domains with open-ended outputs and an existing evaluation ecosystem. Figure 1 presents the complete framework. P2E maps available signals to quality dimensions before exposing human labels and requests structured feedback on semantically distinct disagreements. Human evidence has two roles: it helps locate evaluation gaps and supplies the standard against which candidate mechanisms are selected. Within P2E, ERA is our method for evidence-guided evaluator evolution. With the base model fixed, it evolves how an

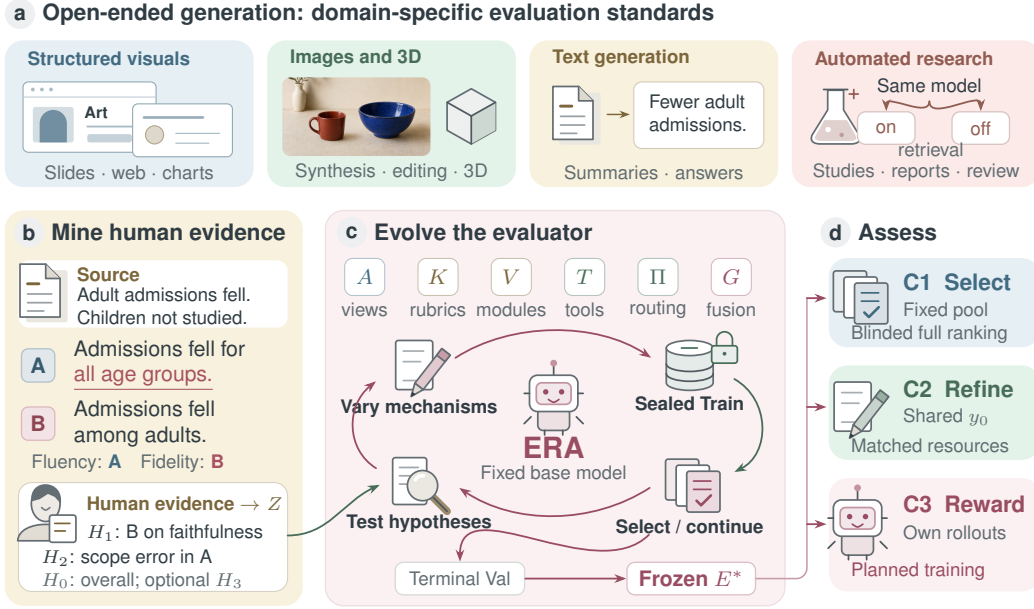


Figure 1: P2E connects domain-specific evaluation to human feedback, evaluator evolution, and independent assessment. (a) Four artifact families. (b) Disagreement and structured human evidence. (c) ERA evolves a six-component program with a fixed base model. (d) C1 selection, C2 refinement, and planned C3 reward training. Artifacts, preferences, and feedback are constructed illustrations; the photograph is generated with GPT Image 2. Arrows denote information flow, not measured improvements.

executable program represents artifacts, acquires observations, and turns them into judgments. The object of learning is the evaluation procedure itself.

Two hypotheses motivate the design. First, disagreements between reliable, nonredundant signals may make a fixed annotation budget more informative than uniform sampling. Second, failures that depend on acquiring, routing, and interpreting evidence may require coordinated evolution of those components. A new observation cannot affect a judgment unless it reaches a module able to interpret it. ERA therefore organizes variation around mechanism hypotheses while selecting complete programs. A failed implementation need not refute its hypothesis, and continuation need not promote the current program. These distinctions motivate matched-budget comparisons with independent proposals and component-restricted search.

A common evolution procedure is instantiated separately for each evaluation standard. We distinguish agreement and selection on fixed candidates (C1), refinement through a frozen critic (C2), and training under a frozen reward (C3). Success at one level does not establish success at the next. This paper develops the feedback-mining formulation, ERA evolution procedure, and controlled studies needed to test these hypotheses.

2 RELATED WORK

Evaluators and feedback acquisition. G-Eval and MT-Bench study model-based judgment and human agreement (Liu et al., 2023; Zheng et al., 2023). Prometheus, Prometheus 2, and Self-Taught Evaluators improve judges through rubric-conditioned or synthetic supervision (Kim et al., 2024a;b; Wang et al., 2024). EvalGen emphasizes aligning the criteria themselves with human preferences, while RewardBench and JudgeBench assess evaluators on challenging comparisons (Shankar et al., 2024; Lambert et al., 2024; Tan et al., 2025). Disagreement-based acquisition has a long history in query-by-committee (Seung et al., 1992). P2E applies this principle to an ecosystem whose signals need not measure the same construct. Its semantic mapping distinguishes same-dimension conflict

from trade-offs and records source dependence. Dimension-wise human judgments can then locate discrepancies that an overall label alone does not resolve.

Optimizing language-model programs. DSPy exposes compositional pipelines to optimization (Khatab et al., 2024). ProTeGi, OPRO, TextGrad, and GEPA use model-generated feedback or proposals to improve instructions and programs (Pryzant et al., 2023; Yang et al., 2024; Yuksekgonul et al., 2025; Agrawal et al., 2026). ADAS, AFlow, and the Darwin Gödel Machine investigate automated construction or evolution of agentic systems (Hu et al., 2025; Zhang et al., 2025b;a). Their search spaces provide relevant controls, rather than a basis for assuming that prior methods cannot change structure. ERA specializes the representation and evidence interface to evaluation, linking ranking failures to observations, criteria, and synthesis while separating an implementation outcome from its underlying direction hypothesis.

Domain quality and downstream use. SummEval and BERTScore address summarization evaluation and text similarity (Fabbri et al., 2021; Zhang et al., 2020); ImageReward, Pick-a-Pic, and GenEval capture different aspects of image preference and alignment (Xu et al., 2023; Kirstain et al., 2023; Ghosh et al., 2023). These signals are useful evidence, not interchangeable definitions of quality. ReAct, Reflexion, and Self-Refine connect observations, reflection, and revision (Yao et al., 2023; Shinn et al., 2023; Madaan et al., 2023). Human-feedback summarization, instruction tuning, and direct preference optimization connect evaluation to model learning (Stiennon et al., 2020; Ouyang et al., 2022; Rafailov et al., 2023). P2E therefore separates fixed-pool evaluation, critic-guided refinement, and reward-guided training as distinct tests of an evolved evaluator.

3 SCOPE AND PROBLEM FORMULATION

A *domain* is a collection of tasks sharing core quality dimensions, their priorities, the overall decision rule, and the human annotation protocol. An artifact type does not define a domain: screenshot-conditioned webpage reconstruction and specification-conditioned webpage generation require different standards. A change of source distribution that preserves the standard is instead a within-domain distribution shift. Appendix A enumerates the full research scope of 22 domains in four families, without assuming that all are ready for experiments. Closed tasks whose complete quality criterion is an executable postcondition fall outside this scope.

Let x be a task with permitted context, y a candidate output, and $\mathcal{Q}_D = \{q_1, \dots, q_m\}$ the quality dimensions for domain D . A human comparison has the core form

$$h = (x, y_a, y_b, q, p, a, \nu), \quad (1)$$

where q is a dimension or overall quality, p is a preference or tie, a records annotator agreement, and ν records provenance. Feedback has four levels: H_0 gives overall preference; H_1 gives dimension-wise preference; H_2 identifies error type, severity, and location; H_3 optionally supplies a free-form rationale. H_1 is central to the dimension-guided design. An H_0 -only corpus remains usable, but it supplies neither human dimensional judgments nor human explanations.

The learned object is an executable evaluation program

$$E_\theta = (A, K, V, T, \Pi, G), \quad (s_\theta, \tau_\theta) = E_\theta(x, y), \quad (2)$$

comprising artifact adapters A , dimension schema and rubrics K , evaluator modules V , tools T , evidence routing Π , and fusion G . The score s_θ is accompanied by an execution trace τ_θ . The base model is fixed outside θ . At deployment, the evaluator receives one candidate and its permitted evidence; opponents, human labels, generator identity, split labels, and mining strata are excluded. We seek agreement with unseen human preferences at an explicitly reported evaluation cost, rather than a single judge shared across all domains.

4 EVALUATOR-GAP-DRIVEN PREFERENCE MINING

4.1 AN EVALUATION SEMANTIC SPACE

Feedback acquisition begins with a versioned inventory of evaluation knowledge, not with a convenient labeled dataset. For each domain, we record dimension definitions, human protocols, rubrics,

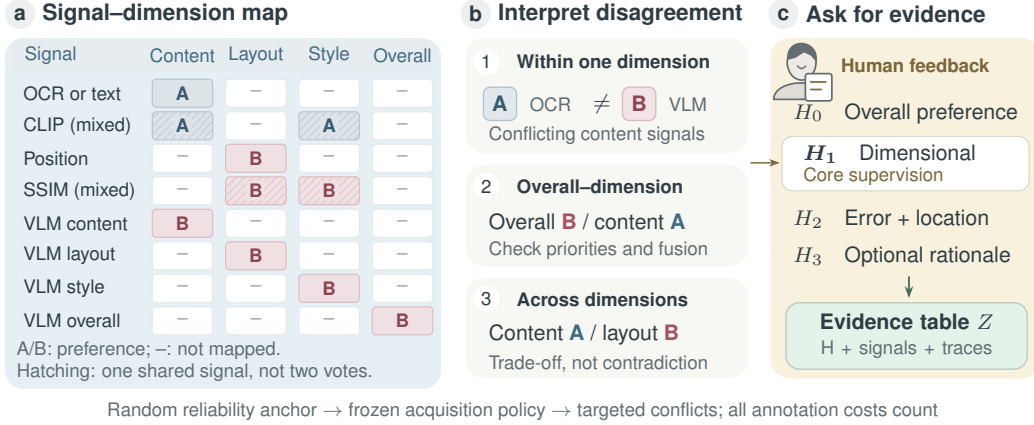


Figure 2: From evaluator signals to informative feedback. Panel (a) is a synthetic webpage example of the many-to-many signal-dimension mapping; letters indicate candidate preference, not scores. Hatched cells share one mixed signal. Panel (b) distinguishes within-dimension contradiction, overall-dimension disagreement, and a trade-off between different dimensions. Panel (c) connects these strata to H_0 – H_3 and the adaptation evidence table Z . The matrix is illustrative and does not report measured scorer behavior.

published metrics, learned scorers, LLM/VLM judges, and external analysis tools. Each source has an input contract, provenance, known limitations, and an execution status. A coverage audit checks the resulting dimension-source matrix. Every core dimension needs executable evidence; conflict mining ordinarily needs two sources with different mechanisms. Several scorers sharing a model or training labels do not establish independent coverage.

The unit of comparison is an *output signal*, rather than an evaluator name. We freeze a many-to-many mapping M_{jq} from signal j to dimension q using its definition and measurement procedure before reading preference labels. A mixed score may connect to several dimensions, but remains one shared signal. Uninterpretable outputs are not assigned dimensions from post hoc correlations. Existing executable metrics can enter the candidate tool library. Metrics reserved for final independent assessment are frozen separately and never enter tools, routing, critique, or reward.

4.2 SEMANTIC DISAGREEMENT AND STRUCTURED ANNOTATION

Raw scores on incompatible scales are not directly subtracted. For a candidate pair $r = (x, y_a, y_b)$, each signal supplies a direction $u_j(r) \in \{-1, 0, +1, \perp\}$, with a normalized margin, repeated-run variance, confidence, and availability. Zero denotes a tie and $\perp$ unavailable or abstaining evidence. A small random anchor estimates reliability and human disagreement before targeted sampling. For the reliable, nonabstaining signal pairs $\mathcal{P}_q(r)$ mapped to the same dimension, a conflict score is

$$c_q(r) = \frac{\sum_{(j,k) \in \mathcal{P}_q(r)} w_{jkq} \mathbb{I}[u_j(r) \neq u_k(r)]}{\sum_{(j,k) \in \mathcal{P}_q(r)} w_{jkq}}, \quad (3)$$

where nonnegative weights reflect anchor reliability and cap the influence of dependent sources. An empty denominator gives no conflict estimate. This formulation leaves the calibration and dependence rule to be fixed before the sampling comparison; disagreement is not itself a correctness label.

We sample three separately recorded semantic strata (Figure 2b). *Within-dimension disagreement* occurs when reliable signals disagree about the same quality criterion. *Overall-dimension disagreement* indicates a possible mismatch in priorities or aggregation. *Cross-dimensional trade-offs* identify cases where different criteria favor different candidates; they are not contradictions between measurements of the same construct. Model differences, repeated-run instability, and small margins further describe each stratum without replacing its semantic definition. Candidate groups cover generators, decoding choices, quality ranges, and failure types. After anchor calibration, the remaining

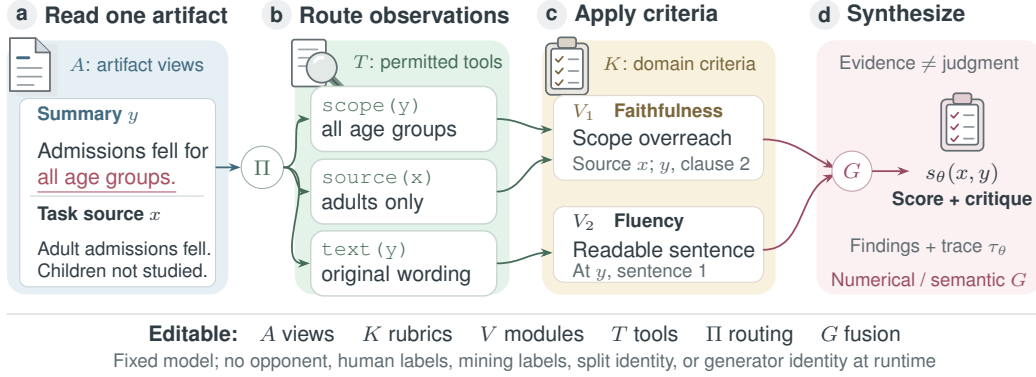


Figure 3: The six-component evaluation program. (a) Read one candidate and permitted task evidence. (b) Route tool observations. (c) Apply dimensional criteria. (d) Fuse sourced findings into a score and critique. The lower rail identifies the editable components. Human feedback supervises evolution, never the deployed scorer. Evidence and judgment remain separate; the summary and findings are a constructed example.

pool’s labels stay hidden until the sampling policy is fixed. Only evolution data are conflict-sampled; final evaluation uses prespecified random or stratified sampling.

4.3 THE ADAPTATION EVIDENCE TABLE

Annotation targets overall and dimensional preferences, with localized diagnostics on a conflict or failure subset. All candidates derived from one input remain grouped. The adaptation evidence table $Z(r, q)$ joins artifact provenance, human judgments and agreement, every source’s direction and uncertainty, evaluator decisions, selected routes, tool evidence, and execution cost. This is the interface from preference mining to ERA. Existing rankings, pointwise ratings, and error annotations require validated adapters and explicit provenance; input/reference pairs alone provide no human supervision. Appendix B specifies the feedback and readiness audits. Neither model-generated explanations nor labels derived from a different quality construct are promoted to human H_1 or H_2 .

5 ERA: EVIDENCE-GUIDED EVALUATOR EVOLUTION

5.1 AN EXECUTABLE EVALUATOR

ERA treats the evaluator’s observation and judgment mechanisms as the objects of evolution. Artifact adapters determine what the program can inspect, from rendered views and source text to spatial structure or interaction traces. Routing selects tools and sends their outputs to evaluator modules with explicit responsibilities. The modules apply the rubric and return findings with provenance and uncertainty; fusion turns these findings into a score and critique (Figure 3). These components are complementary: an available tool changes the judgment only if its evidence is acquired, routed, and interpreted. Existing metrics can serve as tools when their input contracts are satisfied, without requiring every tool to run on every artifact.

Fusion may use a declared numerical rule or semantic synthesis. Semantic synthesis reads specialist findings and their anchored quality scales, groups overlapping or conflicting factors, and locates overall quality on its own scale. It does not average raw scores with different meanings. References to findings must be accounted for, although correct reference bookkeeping does not establish semantic correctness. Appendix E illustrates this distinction.

5.2 MECHANISM HYPOTHESES GUIDE PROGRAM VARIATION

An evolution direction is a hypothesis about why the current program disagrees with human judgment. A summary evaluator, for example, may check individual facts without comparing their scope

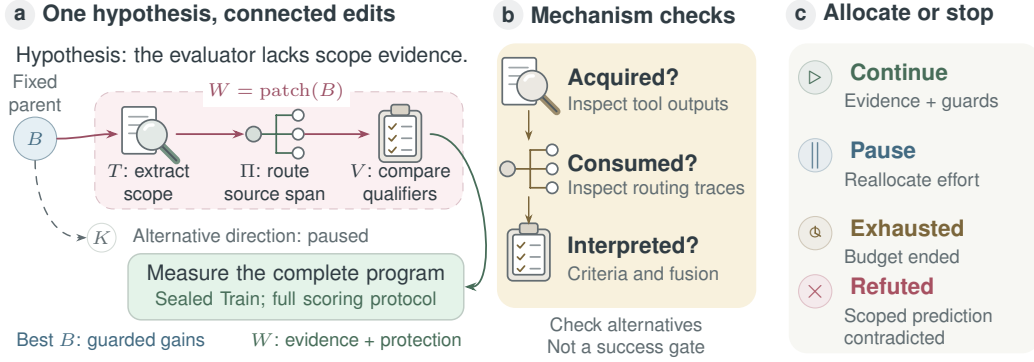


Figure 4: A direction is a hypothesis, not an edit. (a) Connected tool, routing, and interpretation changes share a fixed parent. Best and working programs remain distinct; promotion and continuation both require complete measurement. (b) Check evidence acquisition, consumption, and interpretation. (c) Distinguish continuation, pause, exhaustion, and scoped refutation. The evolution branches are schematic, not an observed optimization trajectory.

against the source. The direction predicts that making source qualifications available to the faithfulness module will change its ordering on scope errors while preserving judgments on faithful summaries. Original training artifacts and repeated traces ground the hypothesis, its competing explanations, target and protection cases, and editable components.

The hypothesis is the unit of exploration; the complete program is the unit of selection. Each elementary edit changes an attributable component of (A, K, V, T, Π, G) . A mechanism that needs a new tool, a route, and a consumer change can be instantiated as one connected candidate with a fixed parent. The complete candidate is measured once as a system; its outcome is not credited separately to every edit. Edit-size and component constraints bound variation, but do not guarantee a small semantic change. Modules may be split, merged, or removed as their responsibilities evolve.

5.3 SELECTION AND EVIDENCE-GUIDED CONTINUATION

ERA separates proposal evidence from candidate selection. The proposal subset $\mathcal{D}_{\text{prop}}$ exposes authorized training artifacts, available human feedback, and versioned traces. A source-disjoint, sealed training subset $\mathcal{D}_{\text{dev}}$ supplies aggregate development measurements only. Local checks on $\mathcal{D}_{\text{prop}}$ assess predictions and regressions without estimating population performance. A frozen candidate is also measured on the common $\mathcal{D}_{\text{dev}}$ cohort. Validation $\mathcal{D}_{\text{val}}$ is terminal; Test/OOD remain locked until the method and analysis are frozen (Appendix C).

Search retains both a best development program and a working program for the active direction (Figure 4a). New directions begin at the best program. Continuing a direction may retain a fully measured intermediate program when new evidence supports further evolution and the protection conditions hold. Development promotion requires improved agreement, complete execution, non-decreasing coverage, and a prespecified stratum-regression guard. It is a selection rule, not a significance test.

Allocation can request evidence, continue, pause, or switch direction. Resource exhaustion is distinct from refutation. A direction is refuted only within a declared scope when evidence contradicts its prediction and relevant alternatives, including missing observations, unused tools, and incorrect fusion, have been examined. Failure of one candidate or a nonsignificant interval is insufficient. On normal completion, an eligible shortlist is frozen from training evidence before terminal validation. System confirmation uses fresh repeated scoring; component attribution is a separate controlled comparison. Neither substitutes for independent Test/OOD evaluation. Appendix D gives Algorithm 1 and the attribution contract.

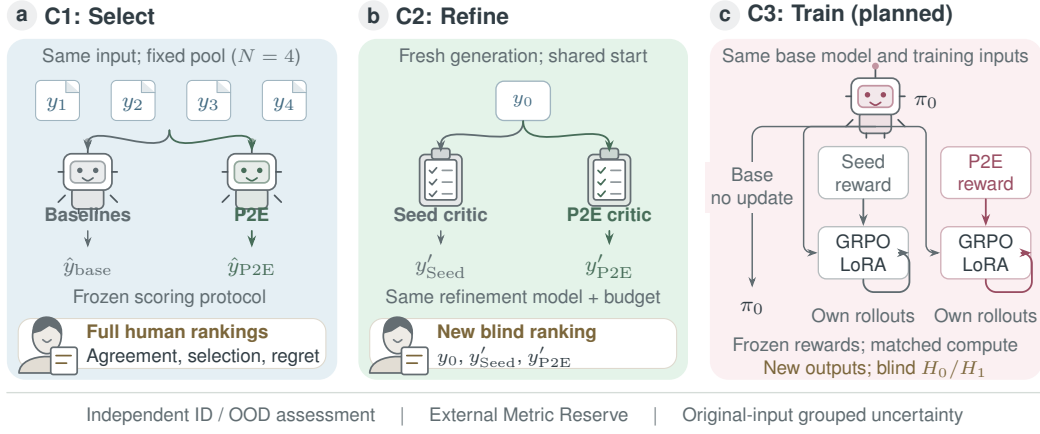


Figure 5: Three controlled uses of the evolved evaluator. Panel (a), C1, holds the candidate pool fixed and compares ranking and selection. Panel (b), C2, branches from a freshly generated shared y_0 and changes only the critic. Panel (c), C3, contrasts an unchanged Base policy with two policies trained under frozen rewards. The trained policies use matched inputs and compute but generate their own rollouts. Final judgments are blinded and external metrics are reserved from evolution. C3 is a planned training study, not an executed experiment.

5.4 POINTWISE MEASUREMENT

Each artifact receives $R \geq 3$ complete scores under a fixed protocol. Aggregation precedes comparison:

$$\bar{s}_\theta(x, y) = \text{median}_{r=1, \dots, R} s_\theta^{(r)}(x, y), \quad z_i = \mathbb{I}[\bar{s}_\theta(x_i, y_i^+) > \bar{s}_\theta(x_i, y_i^-)]. \quad (4)$$

Here y_i^+ is the human-preferred artifact, a designation never exposed to the scorer. Predicted ties are incorrect on strict-preference pairs; human ties require a separate prespecified policy. This is not majority voting over pairwise decisions. Missing executions retain their denominators and are reported through coverage and missing-outcome bounds (Appendix F).

6 EXPERIMENTAL DESIGN

The evaluation has three parallel parts: core results, ablation studies, and analysis. Domains enter confirmatory experiments only after a readiness audit. The complete domain panorama is retained independently of the selected execution scope. Models, datasets, human protocols, budgets, and primary contrasts are frozen before final outcomes are observed.

6.1 CORE RESULTS

C1: evaluator alignment and selection. Frozen P2E evaluators (Figure 5a) are compared with the best single metric, a metric ensemble, a strong static judge, and a static tool-augmented judge. Prompt-optimization and program-search controls use declared versions and comparable edit interfaces. Matched comparisons share the base model, human supervision, inputs, candidate pools, and scoring protocol. Search cost and deployment cost are reported separately.

For each test input, all evaluators receive the same candidate group, with a default design of $N = 4$. Complete human rankings support global pairwise agreement, top-region agreement involving the human top-two candidates, Best-of- N selection, and selection regret. When only ranks are available, normalized regret is $(\text{rank}(\hat{y}) - 1)/(N - 1)$ for the selected output $\hat{y}$. A pair-only dataset supports pairwise evaluation, not an N -way selection claim. Core H_1 , worst-stratum performance, coverage, variance, and cost accompany overall agreement.

ID and OOD results are reported separately. OOD changes dataset or source distribution while preserving the evaluation standard. A difficult subset alone is a difficulty test, not an OOD source. Final sampling is independent of mining conflicts. All splitting and statistical resampling preserve

Table 1: Controlled ablation families. Each row tests a different hypothesis; none is a completed result. The full condition matrix is in Appendix G.

Variable	Principal conditions	Fixed quantities
Preference mining	Random; uncertainty; model conflict; instability; P2E	Human budget, evolution procedure, test set
Feedback detail	$H_0; +H_1; +H_2; +H_3$	Same pairs and annotation provenance
Editable components	Full; rubric/procedure only; fixed tools/routes; routing only	Seed, model, opportunity and measurement budgets
Search allocation	Independent proposals; history only; direction continuation; pause/resume	Initial state, evidence access, total compute
Component attribution	Tool on/off; evolved/seed tool; adaptive/static route; fixed fusion	Other components and paired input groups

original input/source groups. Paired effects use grouped 95% intervals. Existing annotations are reused only when their construct and candidate groups match the endpoint.

C2: critic-guided refinement. For each unseen input, a frozen generator produces one fresh output y_0 . The seed critic and P2E critic each inspect that same output and permitted evidence, then return the same critique format. A common generator performs one refinement step under identical prompts, decoding, tool access, and token budgets (Figure 5b). A separate refinement-validation set determines these settings before ID/OOD generation. New blinded human rankings assess y_0 , Seed-refined, and P2E-refined. The primary contrast is P2E-refined versus Seed-refined, accompanied by improvement and regression rates against y_0 , core dimensional judgments, and reserved external metrics. Labels on pre-existing outputs cannot assess these new outputs.

C3: reward-guided training. The planned training study compares Base, Seed-GRPO, and P2E-GRPO from the same initial checkpoint (Figure 5c), using LoRA and group relative policy optimization (Hu et al., 2022; Shao et al., 2024). The two training arms share prompts and their order, optimizer settings, group size, token limits, update count, and policy-training compute. Rewards are frozen before training. Rollouts are sampled online by each arm’s own evolving policy and are not shared. Reward-call counts and token limits are matched; actual scoring cost is reported separately. Frozen final checkpoints generate new ID/OOD outputs for new blinded H_0/H_1 and reserved external metrics. Checkpoint selection, invalid-output policy, and reward-hacking diagnostics are specified in Appendix H. C3 remains a design-only study here; no parameter training or training benefit is reported.

6.2 ABLATION STUDIES

The ablations separate annotation choices from changes to the evaluator and search procedure (Table 1). Mining strategies are compared at equal human budgets $N_H \in \{20, 50, 100, 200\}$, measured in original-input annotation units. The endpoint is final held-out alignment after evolution, not the fraction of errors found during acquisition. When feedback granularity changes annotation time, both unit count and time/cost are reported. Component controls preserve unaffected interfaces; an invalid dependency is reported instead of silently reverting the entire evaluator to its seed.

6.3 ANALYSIS AND STUDIES

Trajectory analysis reports accepted directions, implementation attempts, resource use, and the evidence supporting each scoped refutation. Failure-regime analysis compares rubric, tool, routing, and fusion contributions with the remaining gap to human agreement. A domain is described as specification-limited, observation-limited, or headroom-limited only after these comparisons; strata within one domain may differ. Feedback-efficiency analysis relates held-out changes to annotation units, time, and granularity. Adapter analysis compares direct human judgments with labels derived from rankings, pointwise scores, or error annotations on the same subset. Repeated scoring and independent search runs are separate sources of variation.

Table 2: C1 reporting template by domain and source cohort. Pair = global pairwise agreement; Top = top-region agreement; BoN = Best-of- N accuracy; Reg = normalized rank regret. All entries are unmeasured; coverage and grouped uncertainty accompany each metric.

Evaluator	ID test				OOD test			
	Pair	Top	BoN	Reg	Pair	Top	BoN	Reg
Best single metric	—	—	—	—	—	—	—	—
Metric ensemble	—	—	—	—	—	—	—	—
Strong static judge	—	—	—	—	—	—	—	—
Static tool-augmented	—	—	—	—	—	—	—	—
Prompt optimization	—	—	—	—	—	—	—	—
Program search	—	—	—	—	—	—	—	—
P2E (ERA)	—	—	—	—	—	—	—	—

Table 3: Downstream contrasts by domain. Report human preference on new outputs with grouped 95% intervals. C3 is planned only; all entries are unmeasured.

Study	Primary contrast	ID preference	OOD preference	Regressions
C2	P2E-refined vs. Seed-refined	—	—	—
C3	P2E-GRPO vs. Seed-GRPO	—	—	—

7 RESULTS

Empirical outcomes are not yet available. Tables 2 and 3 specify per-domain ID/OOD reporting; dashes denote unmeasured quantities, not zero or a method ordering. Appendix I specifies dimensional, reliability, cost, and ablation outputs.

8 DISCUSSION

P2E asks what an evaluator must learn beyond a mapping from outputs to scores. A preference specifies an ordering, but explaining it requires evidence and a standard for interpretation. Mining locates cases where existing signals fail to settle a judgment. ERA then tests hypotheses about what to observe and how those observations should matter, making observation, routing, and interpretation explicit objects of evolution.

Useful exploration need not yield a program worth retaining. Connected changes can instantiate a hypothesis without justifying its continued pursuit. The same distinction applies downstream: correct rankings need not produce actionable critiques, and agreement on fixed candidates need not survive policy optimization. C1, C2, and C3 separate these uses of an evolved evaluator.

8.1 LIMITATIONS AND FUTURE WORK

First, dimensional judgments require additional annotation and construct validation beyond overall preference. Second, disagreement can reflect noise or shared bias, requiring source-dependence audits and matched-budget acquisition studies. Third, interacting components require controlled attribution beyond execution traces. Finally, pointwise scoring assumes a useful within-domain ordering, and repeated development can overfit aggregate feedback. Independent ID/OOD and generation-level studies are needed to test evaluation gains, transfer across standards, and training benefits.

9 CONCLUSION

P2E treats evaluator evolution as learning how to acquire and interpret evidence under a human evaluation standard. Structured preferences guide mechanism hypotheses, while ERA explores implementations and selects complete programs with a fixed base model. Across four domain families, the study design separates alignment from critic-guided refinement and planned reward-guided training, asking when better judgment can also improve generation.

AI USE STATEMENT

Generative AI assisted manuscript drafting, literature organization, and the preparation of editable LaTeX conceptual diagrams. GPT Image 2 generated the photographs in Figures 1a, 6b, and 7b through scene generation and controlled edits. The artifact views and worked examples are synthetic illustrations, not observed cases, human annotations, or performance measurements. No empirical results or parameter-training runs were produced for this draft. Bibliographic identities were checked against primary metadata sources. Author review, responsibility for the final content, and the final disclosure of research-stage AI use remain to be completed before submission.

ETHICS STATEMENT

Human preferences can encode annotator and model biases rather than a universal definition of quality. Domain-specific criteria and uncertainty should remain visible, especially for automated research judgments. Any new annotation study must document consent, compensation, privacy protection, and the applicable institutional review determination before data collection. Artifact licenses and source permissions must be checked independently of technical accessibility. Generated code and evidence tools require restricted execution and controlled external access. Reward-guided optimization requires checks for exploitation of the evaluator and degradation outside its favored criteria. No private artifacts, identifiable annotation records, or experimental traces are included in this working manuscript. Study-specific approvals and release decisions are pending.

REPRODUCIBILITY STATEMENT

Sections 3–6 define semantic mappings, feedback, the evaluator program, evolution, and controlled evaluation. Appendices B–F specify annotation, access boundaries, the algorithm, and measurement. Appendices G–J specify ablations, planned training, reporting outputs, and the experiment manifest. The current manuscript contains no empirical results to reproduce. Before submission, the authors must supply a versioned, appropriately anonymized implementation and permitted data-processing and analysis artifacts with completed manifests. Their availability is not asserted here.

REFERENCES

- Lakshya A Agrawal, Shangyin Tan, Dilara Soylu, Noah Ziemis, Rishi Khare, Krista Opsahl-Ong, Arnav Singhvi, Herumb Shandilya, Michael J Ryan, Meng Jiang, Christopher Potts, Koushik Sen, Alexandros G. Dimakis, Ion Stoica, Dan Klein, Matei Zaharia, and Omar Khattab. GEPA: Reflective Prompt Evolution Can Outperform Reinforcement Learning. In *International Conference on Learning Representations*, 2026. URL <https://arxiv.org/abs/2507.19457>.
- Xin-Sheng Chen, Jiayu Zhu, Pei lin Li, Hanzheng Wang, Shuojin Yang, and Meng-Hao Guo. PresentBench: A Fine-Grained Rubric-Based Benchmark for Slide Generation. *arXiv preprint arXiv:2603.07244*, 2026. URL <https://arxiv.org/abs/2603.07244>.
- Alexander R. Fabbri, Wojciech Kryściński, Bryan McCann, Caiming Xiong, Richard Socher, and Dragomir Radev. SummEval: Re-evaluating Summarization Evaluation. *Transactions of the Association for Computational Linguistics*, 9:391–409, 2021. doi: 10.1162/tacl_a_00373.
- Dhruba Ghosh, Hannaneh Hajishirzi, and Ludwig Schmidt. GenEval: An object-focused framework for evaluating text-to-image alignment. In *Advances in Neural Information Processing Systems*, volume 36, pp. 52132–52152, 2023. doi: 10.52202/075280-2270.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022. URL <https://openreview.net/forum?id=nZeVKeeFYf9>.
- Shengran Hu, Cong Lu, and Jeff Clune. Automated Design of Agentic Systems. In *International Conference on Learning Representations*, 2025. URL <https://arxiv.org/abs/2408.08435>.

- Omar Khattab, Arnav Singhvi, Paridhi Maheshwari, Zhiyuan Zhang, Keshav Santhanam, Sri Vardhamanan, Saiful Haq, Ashutosh Sharma, Thomas T. Joshi, Hanna Moazam, Heather Miller, Matei Zaharia, and Christopher Potts. DSPy: Compiling Declarative Language Model Calls into Self-Improving Pipelines. In *International Conference on Learning Representations*, 2024. URL <https://arxiv.org/abs/2310.03714>.
- Seungone Kim, Jamin Shin, Yejin Cho, Joel Jang, Shayne Longpre, Hwaran Lee, Sangdoo Yun, Seongjin Shin, Sungdong Kim, James Thorne, and Minjoon Seo. Prometheus: Inducing Fine-grained Evaluation Capability in Language Models. In *International Conference on Learning Representations*, 2024a. URL <https://arxiv.org/abs/2310.08491>.
- Seungone Kim, Juyoung Suk, Shayne Longpre, Bill Yuchen Lin, Jamin Shin, Sean Welleck, Graham Neubig, Moontae Lee, Kyungjae Lee, and Minjoon Seo. Prometheus 2: An Open Source Language Model Specialized in Evaluating Other Language Models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pp. 4334–4353, 2024b. doi: 10.18653/v1/2024.emnlp-main.248.
- Yuval Kirstain, Adam Polyak, Uriel Singer, Shahbuland Matiana, Joe Penna, and Omer Levy. Pick-a-Pic: An Open Dataset of User Preferences for Text-to-Image Generation. In *Advances in Neural Information Processing Systems*, volume 36, pp. 36652–36663, 2023. doi: 10.52202/075280-1594.
- Nathan Lambert, Valentina Pyatkin, Jacob Morrison, LJ Miranda, Bill Yuchen Lin, Khyathi Chandu, Nouha Dziri, Sachin Kumar, Tom Zick, Yejin Choi, Noah A. Smith, and Hannaneh Hajishirzi. RewardBench: Evaluating Reward Models for Language Modeling. *arXiv preprint arXiv:2403.13787*, 2024. URL <https://arxiv.org/abs/2403.13787>.
- Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. G-Eval: NLG Evaluation using GPT-4 with Better Human Alignment. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 2511–2522, 2023. doi: 10.18653/v1/2023.emnlp-main.153.
- Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegrefe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. Self-Refine: Iterative Refinement with Self-Feedback. In *Advances in Neural Information Processing Systems*, volume 36, pp. 46534–46594, 2023. doi: 10.52202/075280-2019.
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul F Christiano, Jan Leike, and Ryan Lowe. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, volume 35, pp. 27730–27744, 2022. doi: 10.52202/068431-2011.
- Reid Pryzant, Dan Iter, Jerry Li, Yin Lee, Chenguang Zhu, and Michael Zeng. Automatic Prompt Optimization with “Gradient Descent” and Beam Search. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 7957–7968, 2023. doi: 10.18653/v1/2023.emnlp-main.494.
- Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, and Chelsea Finn. Direct Preference Optimization: Your Language Model is Secretly a Reward Model. In *Advances in Neural Information Processing Systems*, volume 36, pp. 53728–53741, 2023. doi: 10.52202/075280-2338.
- H. S. Seung, M. Oppen, and H. Sompolinsky. Query by committee. In *Proceedings of the Fifth Annual Workshop on Computational Learning Theory*, pp. 287–294. ACM, 1992. doi: 10.1145/130385.130417. URL <https://doi.org/10.1145/130385.130417>.
- Shreya Shankar, J.D. Zamfirescu-Pereira, Bjoern Hartmann, Aditya Parameswaran, and Ian Arawjo. Who Validates the Validators? Aligning LLM-Assisted Evaluation of LLM Outputs with Human Preferences. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*, pp. 1–14, 2024. doi: 10.1145/3654777.3676450.

- Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024. URL <https://arxiv.org/abs/2402.03300>.
- Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 36, pp. 8634–8652, 2023. doi: 10.52202/075280-0377.
- Chenglei Si, Yanzhe Zhang, Ryan Li, Zhengyuan Yang, Ruibo Liu, and Diyi Yang. Design2Code: Benchmarking Multimodal Code Generation for Automated Front-End Engineering. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pp. 3956–3974, 2025. doi: 10.18653/v1/2025.naacl-long.199.
- Nisan Stiennon, Long Ouyang, Jeffrey Wu, Daniel Ziegler, Ryan Lowe, Chelsea Voss, Alec Radford, Dario Amodei, and Paul F Christiano. Learning to summarize with human feedback. In *Advances in Neural Information Processing Systems*, volume 33, pp. 3008–3021, 2020. URL https://proceedings.neurips.cc/paper_files/paper/2020/file/1f89885d556929e98d3ef9b86448f951-Paper.pdf.
- Sijun Tan, Siyuan Zhuang, Kyle Montgomery, William Y. Tang, Alejandro Cuadron, Chenguang Wang, Raluca Ada Popa, and Ion Stoica. JudgeBench: A Benchmark for Evaluating LLM-based Judges. In *International Conference on Learning Representations*, 2025. URL <https://arxiv.org/abs/2410.12784>.
- Tianlu Wang, Ilia Kulikov, Olga Golovneva, Ping Yu, Weizhe Yuan, Jane Dwivedi-Yu, Richard Yuanzhe Pang, Maryam Fazel-Zarandi, Jason Weston, and Xian Li. Self-Taught Evaluators. *arXiv preprint arXiv:2408.02666*, 2024. URL <https://arxiv.org/abs/2408.02666>.
- Jiazheng Xu, Xiao Liu, Yuchen Wu, Yuxuan Tong, Qinkai Li, Ming Ding, Jie Tang, and Yuxiao Dong. ImageReward: Learning and Evaluating Human Preferences for Text-to-Image Generation. In *Advances in Neural Information Processing Systems*, volume 36, pp. 15903–15935, 2023. doi: 10.52202/075280-0700.
- Chengrun Yang, Xuezhi Wang, Yifeng Lu, Hanxiao Liu, Quoc V. Le, Denny Zhou, and Xinyun Chen. Large Language Models as Optimizers. In *International Conference on Learning Representations*, 2024. URL <https://arxiv.org/abs/2309.03409>.
- Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing Reasoning and Acting in Language Models. In *International Conference on Learning Representations*, 2023. URL <https://arxiv.org/abs/2210.03629>.
- Mert Yuksekgonul, Federico Bianchi, Joseph Boen, Sheng Liu, Pan Lu, Zhi Huang, Carlos Guestrin, and James Zou. Optimizing generative AI by backpropagating language model feedback. *Nature*, 639(8055):609–616, 2025. doi: 10.1038/s41586-025-08661-4.
- Jenny Zhang, Shengran Hu, Cong Lu, Robert Lange, and Jeff Clune. Darwin Godel Machine: Open-Ended Evolution of Self-Improving Agents. *arXiv preprint arXiv:2505.22954*, 2025a. URL <https://arxiv.org/abs/2505.22954>.
- Jiayi Zhang, Jinyu Xiang, Zhaoyang Yu, Fengwei Teng, Xiong-Hui Chen, Jiaqi Chen, Mingchen Zhuge, Xin Cheng, Sirui Hong, Jinlin Wang, Bingnan Zheng, Bang Liu, Yuyu Luo, and Chenglin Wu. AFlow: Automating Agentic Workflow Generation. In *The Thirteenth International Conference on Learning Representations*, 2025b. URL <https://openreview.net/forum?id=z5uVAKwmjf>.
- Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q. Weinberger, and Yoav Artzi. BERTScore: Evaluating Text Generation with BERT. In *International Conference on Learning Representations*, 2020. URL <https://arxiv.org/abs/1904.09675>.

Hao Zheng, Xinyan Guan, Hao Kong, Wenkai Zhang, Jia Zheng, Weixiang Zhou, Hongyu Lin, Yaojie Lu, Xianpei Han, and Le Sun. PPTAgent: Generating and Evaluating Presentations Beyond Text-to-Slides. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 14402–14418, 2025. doi: 10.18653/v1/2025.emnlp-main.728.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, Hao Zhang, Joseph Gonzalez, and Ion Stoica. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems*, volume 36, pp. 46595–46623, 2023. doi: 10.52202/075280-2020.

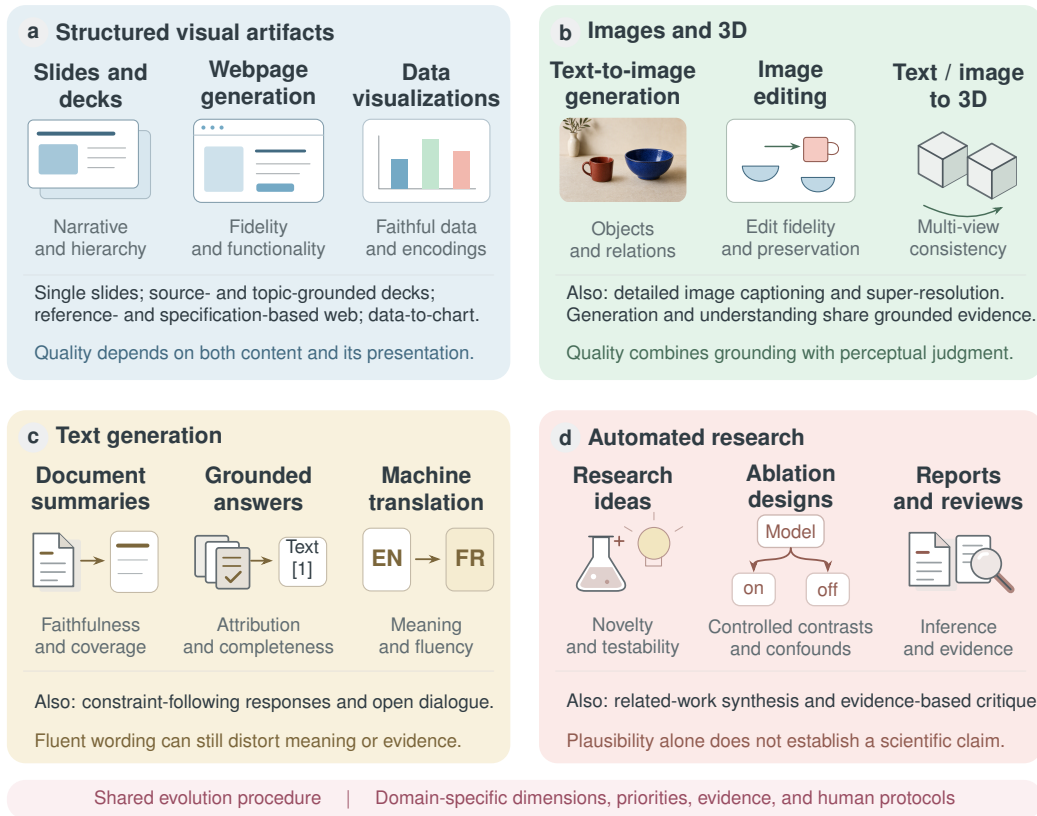


Figure 6: Diverse artifacts, distinct evaluation standards. Representative tasks span (a) structured visual artifacts, (b) images and 3D, (c) text generation, and (d) automated research. Each family combines several quality considerations; the full 22-domain taxonomy is in Table 4. P2E shares an evolution procedure across these standards, not a universal scorer. Artifact views are constructed illustrations, including the GPT Image 2 photograph; they do not indicate benchmark coverage, implemented adapters, or completed experiments.

A DOMAIN PANORAMA AND EVALUATION STANDARDS

The research scope is organized by evaluation standard, not by file format. Table 4 retains all 22 candidate domains. Its dimensions are a starting taxonomy to be refined from domain evaluation knowledge before selecting datasets. Each instantiated domain must additionally fix priorities, decision rules, and an annotation protocol. Inclusion in this panorama does not establish data readiness, an implemented adapter, or empirical transfer. Figure 6 illustrates representative tasks and the quality criteria that distinguish the four families.

Table 4: Complete domain panorama. The four families organize the scope; every row requires its own standard and readiness audit. These are study candidates, not 22 completed evaluations.

ID	Domain	Core quality considerations
Structured visual artifacts		
TD-SV1	Brief-conditioned single-slide generation	Brief coverage, hierarchy, readability, composition, and visual consistency.
TD-SV2	Screenshot-conditioned webpage reconstruction	Reference content, spatial arrangement, typography, and visual fidelity.
TD-SV3	Data-to-visualization generation	Data fidelity, valid encodings, readability, and communicative usefulness.

Table 4 continued

ID	Domain	Core quality considerations
TD-SV4	Specification-conditioned webpage generation	Requirement satisfaction, functional behavior, usability, and visual organization.
TD-SV5	Source-grounded slide-deck generation	Source faithfulness, narrative, coverage, slide readability, and deck consistency.
TD-SV6	Topic-conditioned slide-deck generation	Relevance, factual support, narrative completeness, audience fit, and presentation quality.
Image generation and understanding		
TD-IG1	Detailed image-caption generation	Grounding, coverage, relation accuracy, specificity, and absence of hallucination.
TD-IG2	Text-to-image generation	Prompt alignment, object and relation accuracy, visual coherence, and preference.
TD-IG3	Single-source instruction-based image editing	Instruction fulfillment, preservation of unrequested content, and image quality.
TD-IG4	LR-conditioned image super-resolution	Reference fidelity when available, perceptual detail, artifact avoidance, and plausible texture.
TD-IG5	Text-to-3D object generation	Prompt correspondence, geometry, appearance, and multi-view consistency.
TD-IG6	Image-to-3D object generation	Source-view fidelity, plausible unseen geometry, appearance, and multi-view consistency.
Text generation		
TD-TG1	Source-grounded document summarization	Faithfulness, relevance, coverage, coherence, and concision.
TD-TG2	RAG-supported long-form answer generation	Answer correctness, evidence grounding, attribution, completeness, and organization.
TD-TG3	Machine translation	Meaning preservation, omissions and additions, fluency, terminology, and style.
TD-TG4	Instruction-constraint-following response generation	Constraint satisfaction, substantive correctness, relevance, and response quality.
TD-TG5	Open-ended dialogue-response generation	Contextual appropriateness, helpfulness, consistency, safety, and conversational quality.
Automated research		
TD-AR1	Research-idea generation	Novelty relative to evidence, significance, feasibility, and testability.
TD-AR2	ML/NLP ablation-plan generation	Hypothesis identification, controlled contrasts, confound management, and feasibility.
TD-AR3	Table-to-insight report generation	Numerical faithfulness, appropriate inference, uncertainty, and relevance.
TD-AR4	Related-work generation	Coverage, citation correctness, faithful comparison, and synthesis.
TD-AR5	Peer-review critique generation	Technical accuracy, evidence, significance assessment, fairness, and actionable criticism.

Representative resources include PPTAgent and PresentBench for presentations, Design2Code for screenshot-conditioned webpages, SummEval for summarization, and image-preference and alignment resources (Zheng et al., 2025; Chen et al., 2026; Si et al., 2025; Fabbri et al., 2021; Kirstain et al., 2023; Xu et al., 2023; Ghosh et al., 2023). Their outputs, references, metrics, and annotations have different evidential roles; a resource is not automatically a source of the required human preference. A benchmark collection spanning several standards remains a dataset collection, not an additional domain. Research execution with a complete executable postcondition is excluded, while ideation, study design, interpretation, writing, and critique remain open-ended evaluation problems.

Dataset roles. Dev supports acquisition and evaluator development, ID tests unseen inputs from the development source distribution, and OOD changes source distribution while preserving the standard. External datasets provide separately locked assessment only and do not determine semantic mappings or acquisition. Harder examples from the same source are a difficulty stratum,

Table 5: Permitted uses of existing data. An adapter changes representation, not the quality construct or the provenance of the original judgment.

Source category	Required evidence	Permitted use
Direct human feedback	Aligned input, readable same-input candidates, and original pair or ranking judgments	H_0 when overall quality was judged; H_1/H_2 only if originally annotated
Adapted feedback	Recoverable candidate groups and valid rankings, pointwise scores, or error annotations	Derived pairs with original groups and explicit ranking/pointwise/error provenance
Materials only	Inputs, references, editable artifacts, or metric outputs without matched human judgments	Fresh generation, method development, or independent metrics; not human supervision
New annotation	New outputs or a mismatch between existing labels and the endpoint	Direct judgments under the frozen protocol; required for new C2/C3 outputs

not a new OOD domain. A domain selected for confirmatory execution must pass the audit below; selection does not remove unselected domains from the research scope.

B FEEDBACK PROVENANCE, ANNOTATION, AND READINESS

Data reuse follows the provenance rules in Table 5.

Annotation units and levels. One annotation unit is an original input and its anonymized candidate group, including permitted references. Pairs derived from that group remain dependent. The core tuple in Equation 1 is retained together with dataset and license, source grouping, generator and prompt version, collection time, annotator pool, agreement, and conversion provenance. Generator identity is recorded for audit but hidden from annotators and the deployed scorer.

The planned evolution protocol collects H_0 and core H_1 , with H_2 on conflict and failure subsets. H_3 is optional. Two independent annotators with adjudication may support development; final ID/OOD assessment targets three independent annotators and blinded complete rankings. Core dimensions and the tie policy are fixed in advance. A model or an author decision is not substituted for a missing independent final annotation. Figures 7a–d illustrate the distinctions without supplying study data. An H_0 -only instantiation must be reported as such; the framework’s richer interface does not establish that a run actually received H_1 or H_2 .

Label-blind acquisition. Freeze domain knowledge, signal definitions, semantic mappings, and the calibration recipe before accessing preference labels. Select a random anchor, reveal only its authorized training judgments to estimate reliability, then freeze the targeted sampling policy before opening labels in the remaining pool. Charge anchor annotations to every strategy’s human budget. An existing labeled dataset must simulate the same access order. The final cohort is sampled independently of conflicts and never contributes to reliability calibration.

Source and tool status. Registry entries distinguish executable, planned, and blocked sources (Table 6). A publication or code link without compatible inputs, weights, or dependencies is not an executed signal. Available tools begin as candidates; a route must explicitly activate them. Source independence is assessed at the signal level, including model, training-label, and metric dependencies. Model-only comparisons hold rubric, evidence, context, and reasoning budget fixed. Replacing both the scoring model and its evidence-collection agent instead compares full systems. Deterministic or fixed learned metrics are not duplicated for every judge model.

C DATA ACCESS AND INDEPENDENT ASSESSMENT

Figure 8 separates proposal evidence, adaptive development measurements, and terminal assessment. The scorer always receives the same pointwise input contract, regardless of the cohort in which it is invoked. Only the authorized diagnostic process sees human labels and pair identities.

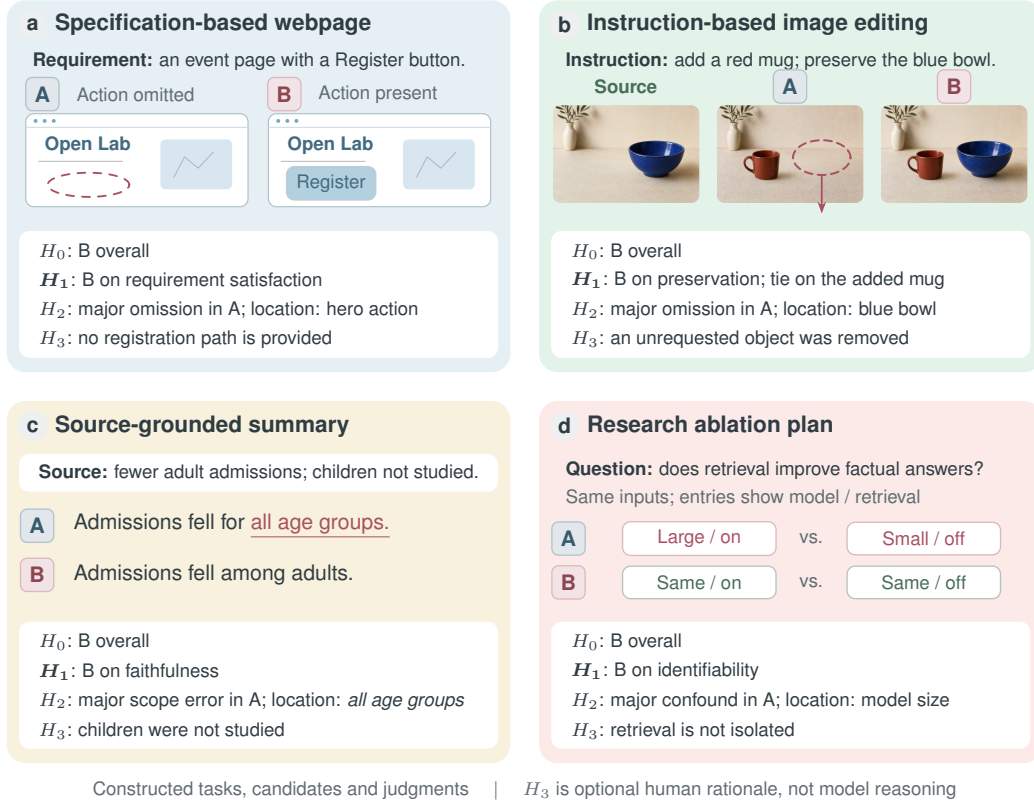


Figure 7: A common feedback schema across distinct quality standards: (a) webpage requirement satisfaction, (b) image-edit preservation, (c) summary faithfulness, and (d) research identifiability. Each example separates overall preference, dimensional judgment, a localized diagnostic, and an optional rationale. All tasks, candidates, and judgments are constructed illustrations, not benchmark cases or collected human feedback. GPT Image 2 generated the source photograph and controlled edits. The labels demonstrate the annotation schema, not agreement rates or evaluator performance.

Training and validation. Proposal-Train evidence may inform evolution directions, candidates, and allocation. Sealed Train reveals prespecified aggregate agreement, coverage, and stratum guards without examples, labels, or per-case errors. Aggregates remain adaptive development evidence, not a final performance estimate. The terminal shortlist and its order are fixed from training evidence before Val access, and every shortlisted outcome is reported. Repeated Val calls can assess stability but do not create an independent test cohort. Interrupted searches do not obtain a new terminal assessment merely because they stopped.

External metrics. The External Metric Reserve is frozen before search. Reserved metrics and their equivalent wrappers are excluded from the executable tool library, router, diagnostician, critique, reward, and checkpoint-selection criteria. If an endpoint was used for optimization, it loses independent-assessment status and must be replaced or explicitly reported as an optimization metric. This restriction does not forbid using other existing metrics as tools.

D ERA ALGORITHM AND ATTRIBUTION CONTRACT

Hypothesis state. Each direction records a mechanism, its evidence, alternative explanations, a falsifiable prediction, protected cases, editable components, and the observations needed to assess the prediction. Local diagnostics are not a mandatory success gate for broader sealed-development measurement. Continuation retains a working program only when its measurement, new evidence,

Table 6: Readiness criteria before a domain enters confirmatory evaluation. These are audit requirements, not completed coverage scores.

Audit object	Required record and exclusion condition
Evaluation standard	Open-endedness, dimensions and hierarchy, priorities, decision rule, and human protocol. Exclude tasks fully decided by an executable postcondition.
Knowledge provenance	Benchmarks, expert guidance, rubrics, and methods with versions, dates, input assumptions, known biases, and availability.
Source coverage	Dimension $\times$ source-type matrix: deterministic, learned metric, LLM/VLM judge, external analysis tool. Each core dimension needs executable evidence.
Source independence	For conflict mining, ordinarily two mechanisms per dimension. Record shared models, training labels, metrics, and mixed signals; multiple names are not independent votes.
Executable artifacts	Recoverable input/candidate groups, readable artifacts, references, permissions, and source fingerprints. Missing assets do not become scorer disagreement.
Conflict pool	Three semantic strata, availability, margins, repeat variance, anchor reliability, dependence caps, and uncovered strata. Noise alone is not a gap.
Feedback fidelity	Direct versus derived provenance, agreement, mapping to $H_0/H_1/H_2$, adapter checks on a common subset, and additional annotation needs.
Source isolation	Original-input deduplication, Dev/ID/OOD/External roles, contamination review, and a pre-search External Metric Reserve.

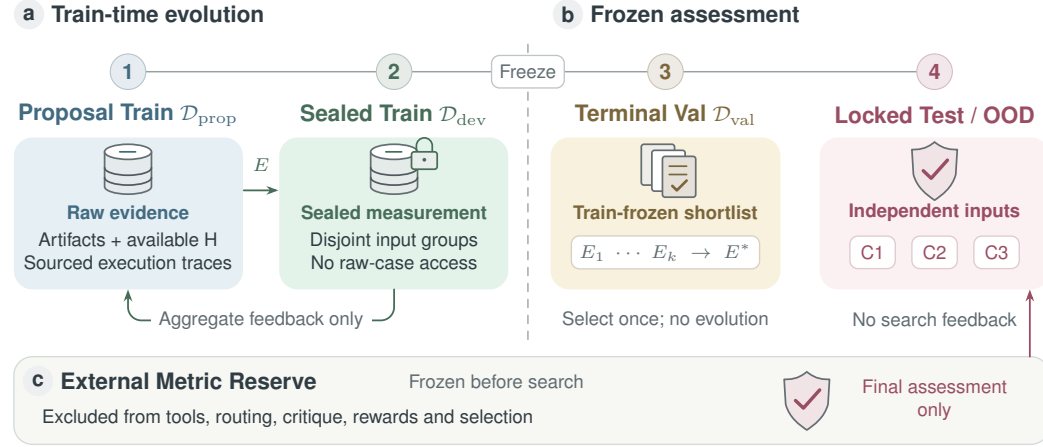


Figure 8: Data and metric boundaries. Panel (a) exposes raw proposal-Train evidence but returns only aggregate measurements from a disjoint sealed Train cohort. Panel (b) assesses a frozen shortlist on Val and the frozen method on locked Test/OOD. Panel (c) reserves metrics for independent final assessment. The reserve cannot be reintroduced through tools, critique, reward, aliases, or checkpoint selection. All partitions preserve original-input/source groups.

and protection conditions justify doing so. The best program is a separate selection state; retaining a working program does not redefine the best score.

Promotion and confirmation. Development promotion requires complete execution, a strict agreement gain, nondecreasing coverage, and a prespecified worst-stratum regression guard. Guard thresholds and tie rules are fixed before search. These are development selection rules rather than statistical significance claims. Normal completion is followed by fresh repeated terminal measurements of the frozen shortlist. Selection uncertainty and the number of assessed candidates are reported.

Attribution. An elementary mutation changes one attributable component. A connected patch may introduce a tool, a route, and a consumer change against one fixed parent. The bundle receives one system-level outcome, not three independent causal credits. Component attribution requires paired interventions with all other interfaces held fixed, including tool on/off, evolved versus seed tool, and adaptive versus static routing. If disabling a dependency makes a program invalid, report

Algorithm 1 ERA within the P2E framework

Require: Seed E_0 , proposal evidence Z_{prop} , sealed development interface, fixed budgets and measurement rule

- 1: Measure E_0 on the common sealed Train cohort; set best program $B \leftarrow E_0$
- 2: Initialize direction history and allocation state
- 3: **while** search resources remain **do**
- 4: Choose to request evidence, start, continue, pause, or switch direction
- 5: **if** requesting evidence **then**
- 6: Acquire authorized proposal-Train evidence; charge its cost
- 7: **else if** starting or continuing a direction **then**
- 8: Set parent $P \leftarrow B$ for a new direction, or the eligible working program
- 9: Specify prediction, alternatives, target cases, and protection conditions
- 10: Construct a bounded atomic edit or connected patch $E' \leftarrow \text{patch}(P)$
- 11: Inspect local diagnostics; retain observations and failures
- 12: Measure frozen E' on the common sealed Train cohort
- 13: Record the full attempt, component changes, costs, and aggregate outcome
- 14: **if** measurement is complete and the development promotion rule holds **then**
- 15: $B \leftarrow E'$
- 16: **end if**
- 17: **if** measurement is complete and the continuation rule holds **then**
- 18: Retain E' as the direction's working program
- 19: **end if**
- 20: Update direction status without equating a failed candidate with refutation
- 21: **else**
- 22: Save the direction state and reallocate the remaining resources
- 23: **end if**
- 24: **end while**
- 25: On normal completion, freeze an eligible shortlist using training evidence only
- 26: Assess the shortlist on terminal Val under a prespecified selection rule
- 27: Freeze the final evaluator and analysis before independent Test/OOD assessment

that limitation and use a compatible control where justified. System confirmation and attribution answer different questions.

A scoped refutation records the contradicted prediction and the evidence that rules out relevant implementation alternatives. Missing observations, an unselected tool, incorrect interpretation, or insufficient resources leave different conclusions. Neither one unsuccessful implementation nor a nonsignificant difference alone refutes the direction.

E SEMANTIC SYNTHESIS AND ANCHORED SCORING

Semantic synthesis is an optional fusion procedure, compared with a declared numerical rule. A specialist report contains a finding identifier, supporting evidence, a dimension, uncertainty, and an anchored quality judgment. The synthesizer groups findings into factors while preserving overlap and contradiction (Figure 9a–b). Duplicate descriptions of the same defect should not become independent penalties, and a favorable style judgment should not silently cancel a factual error when the domain standard prioritizes faithfulness.

Let $a_k < a_{k+1}$ be adjacent declared anchors and $\lambda \in [0, 1]$ locate quality between them. The resulting scalar is

$$s = (1 - \lambda)a_k + \lambda a_{k+1}. \quad (5)$$

The anchors have domain-specific verbal meanings. Finding-level scores with different meanings are not averaged merely because they are numeric. Reference completeness, semantic correctness, and calibration are separate properties and require separate checks. Figure 9c illustrates only the arithmetic, using $a_k = 50$, $a_{k+1} = 75$, and $\lambda = 0.4$.

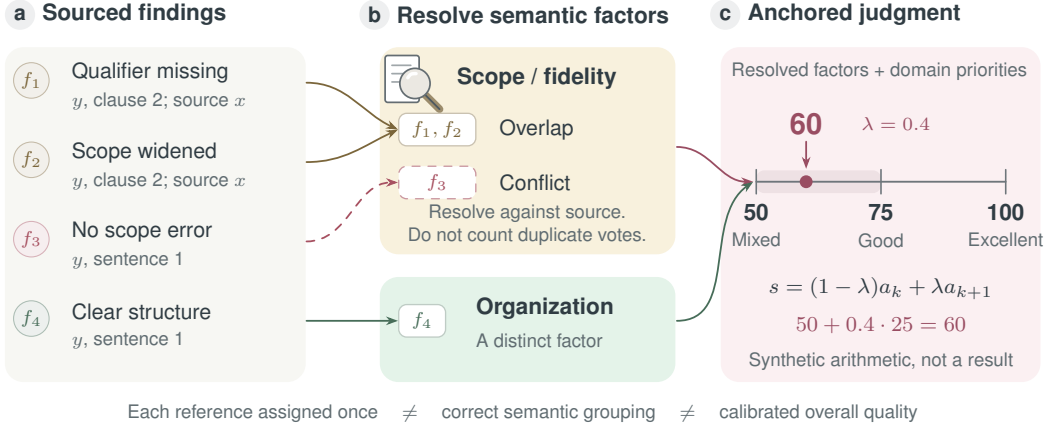


Figure 9: Optional semantic synthesis. Panel (a) supplies sourced findings; (b) groups overlapping and conflicting statements into factors; (c) locates overall quality between prespecified anchors. Every finding reference is assigned once, but bookkeeping alone cannot validate the grouping or the quality judgment. The anchor values and interpolation are a synthetic arithmetic example, not an evaluator result or a universal scale for all domains.

F MEASUREMENT AND STATISTICAL ANALYSIS

Completeness and denominators. For a prespecified strict-preference cohort of N pairs, let $c_i = 1$ when both artifacts have all R valid complete scores. Let $m = \sum_i c_i$ and $k = \sum_{i:c_i=1} z_i$, with z_i from Equation 4. Report completed-pair accuracy k/m when $m > 0$, coverage m/N , and the missing-outcome interval

$$\left[\frac{k}{N}, \frac{k + N - m}{N} \right]. \quad (6)$$

This interval bounds full-cohort agreement under arbitrary missing outcomes; it is not a statistical confidence interval. A missing repeat is not replaced by the median of surviving calls. Completeness includes selected routes and tools, not only a final score. A valid observation that an artifact fails a quality check differs from an execution error.

Ranking and selection endpoints. For the default $N = 4$ design, human complete rankings support all strict pairs. Global agreement averages within input groups before aggregation across inputs. Top-region agreement uses prespecified strict comparisons involving at least one human top-two candidate. Best-of- N measures selection of a human top-ranked candidate; normalized rank regret is defined in Section 6.1. Human ties, evaluator ties, and incomplete rankings are governed by a frozen policy and reported separately. Pair-only data do not supply missing ranks or cardinal utility. Core H_1 agreement is reported by dimension before any macro average.

Randomness and repeated use. Freeze $R \geq 3$, score precision, retry policy, artifact reuse, and cache identity. Retries are outcome-blind and uniformly applied. Repeated scores are not independent task observations. Paired uncertainty uses original-input/source group resampling, preserving candidate groups and comparisons; the planned interval level is 95%. Report independent search seeds separately from scoring repeats, and distinguish within-run execution variation from between-search variation. Retain failed, rejected, incomplete, and interrupted attempts in the authorized analysis, without favorable resampling or denominator changes.

C2/C3 human outcomes. Report wins, losses, ties, invalid outputs, and major regressions per input group. A prespecified tie-aware preference probability may assign half credit to a tie, with its denominator and missingness bounds stated. Report C2 changes against the shared y_0 in addition to the critic contrast, and C3 changes against Base in addition to the reward contrast. Intervals and costs accompany effect sizes. Multiple domains, dimensions, and ablations require declared

Table 7: Planned ablation conditions and the inference each comparison supports. All conditions require matched evidence access and accounted computation. An oracle condition is diagnostic, not a deployable comparator.

Factor	Conditions	Endpoint or interpretation
Acquisition	Random; judge uncertainty; cross-model conflict; run instability; P2E semantic multi-source disagreement	Final held-out alignment at equal $N_H \in \{20, 50, 100, 200\}$; acquisition yield alone is insufficient.
Human detail	Same pairs with H_0 , $H_0 + H_1$, $H_0 + H_1 + H_2$, and $H_0 + H_1 + H_2 + H_3$	Marginal value of granularity; report annotation units and annotation time.
Editable scope	Full (A, K, V, T, Π, G) ; rubric/procedure only (K, V) ; fixed tools and static routes; routing only	Whether additional observations or structural changes are needed beyond criteria changes.
Evidence ceiling	Oracle evidence and/or oracle routing with access declared	Diagnostic headroom under privileged information; no deployment claim.
Search allocation	Independent one-shot proposals; history-only proposals; direction depth-first; direction search with pause/resume	Value of direction management at equal total search and measurement budgets.
Tool activation	Tool on/off with compatible consumers	Contribution of acquiring the observation, not the whole program bundle.
Tool implementation	Evolved tool versus seed tool	Contribution of changed measurement or evidence extraction.
Evidence route	Adaptive versus fixed routing with the same tool set	Contribution of which observations reach which modules.
Rubric and modules	Evolved versus fixed criteria/procedures	Contribution of judgment changes with observations held fixed.
Fusion	Declared numerical rule versus semantic synthesis	Contribution of resolving priorities, overlap, and conflicting findings.
Feedback adapter	Direct pairs versus derived ranking/pointwise/error labels on the same subset	Label fidelity and its downstream effect; keep original input dependence.

primary contrasts and a prespecified multiplicity policy. A search-selected development maximum is not substituted for any final endpoint.

G FULL ABLATION MATRIX

The comparisons in Table 7 freeze the initial program, evidence access, edit interfaces, proposal opportunities, and measurement budgets. All diagnostic and validation calls are charged, including unsuccessful attempts. If method interfaces differ, report that difference instead of attributing a resource advantage to the search policy. Attribution uses paired original-input groups and pre-specified compatible component replacements. Observed interactions may limit a one-component interpretation.

H PLANNED C3 TRAINING PROTOCOL

C3 is specified to test reward usefulness independently of evaluator agreement and critic usefulness. It is not an executed experiment, and no parameter training is part of this manuscript preparation.

Under the controls in Table 8, the frozen reward produces scalar scores within each rollout group using the same normalization and advantage rule. Zero-variance groups, invalid artifacts, and execution failures require a common prespecified policy. Deterministic checks may enforce format or executability, but do not define the entire reward for open-ended quality.

The pilot uses development validation only. If its prespecified stability or reward-hacking gate fails, expansion stops for that domain and the failure is retained. Success elsewhere is not a substitute.

Table 8: Controls for the planned LoRA + GRPO comparison. Numerical hyperparameters, pilot thresholds, and final sample sizes remain to be frozen in the experiment manifest.

Control	Contract
Arms	Base is untrained; Seed-GRPO and P2E-GRPO start from the identical base checkpoint.
Training data	Same unlabeled training inputs and order; source-level deduplication and contamination audit.
Policy updates	Same LoRA structure, optimizer, learning rate, update count, group size, KL rule, checkpoint cadence, and policy-training compute.
Online samples	Each updated policy samples its own rollout groups. Do not share trajectories once policies diverge.
Reward version	Freeze program, prompt, tools, routing, aggregation, normalization, ties, abstention, and invalid-output policy before training.
Reward resources	Equal reward-call counts and token caps; report actual inference time, tool use, and monetary or compute cost separately.
Training validation	Input-disjoint development validation selects checkpoints and checks reward exploitation; final metrics and final human labels remain locked.
Pilot gate	Prespecified acceptable validity, reward variance, KL, length, stability, held-out quality, and reward-hacking diagnostics before expansion.
Final generation	Freeze checkpoints and decoding, then generate new Base, Seed-GRPO, and P2E-GRPO outputs on the same unseen ID/OOD inputs.
Final judgment	New blinded H_0 and core H_1 , reserved external metrics, invalid/major-regression rates, length and format shifts, and total cost.

Final sample sizes are determined from pilot variance and a declared minimum detectable effect before test generation. Test inputs and judgments do not participate in training, checkpoint selection, reward design, or failure-driven edits. Any deliberate sharing of locked input pools across C1/C2/C3 must be prespecified with an access audit; their newly generated outputs and judgments are never treated as reusable old labels.

I REQUIRED RESULT AND ANALYSIS OUTPUTS

Empirical plots for the outcomes in Table 9 require frozen reporting cohorts and analysis. Suitable outputs include human-budget curves, per-domain component effects, annotated search trajectories, and C2/C3 regression breakdowns. The present figures are conceptual diagrams and synthetic worked examples, not provisional substitutes for those measurements.

J EXPERIMENT MANIFEST

Before submission, complete Table 10, replace result placeholders with permitted source-traceable measurements, finalize the empirical scope without changing the method’s definition of a domain, and supply appropriately anonymized reproducibility artifacts. Unmeasured hypotheses remain hypotheses.

Table 9: Companion outputs to the core result templates. Every empirical output requires authorized frozen data and a traceable analysis; none is supplied by the conceptual illustrations.

Output	Required summary	Interpretation limit
C1 alignment	Global/top-region agreement, per-dimension H_1 , worst stratum, ID/OOD, coverage, repeated-score variation	Pairwise improvement does not establish selection or generation gains.
C1 selection	Best-of- N , normalized rank regret, complete human rankings, grouped intervals	Rank regret is not cardinal utility; pair-only corpora are insufficient.
C2 refinement	New blinded contrast, changes against y_0 , major regressions, dimensional quality, external metrics	Fresh shared initial outputs and matched refinement are necessary.
C3 training	New blinded reward contrast, Base sanity check, external metrics, validity, reward-hacking diagnostics	Design-only until separately authorized training and final evaluation occur.
Efficiency	Human units and time, search calls/tokens/tool use, runtime, deployment and training costs	More compute or annotation detail is not a free method gain.
Trajectory	Directions, implementation attempts, continuation/pause, exhaustion/refutation, resource allocation	A schematic tree is not an optimization trace or evidence of efficiency.
Mechanism	Paired component effects, interactions, remaining gap to human agreement, scoped counterexamples	Assign limitation regimes only after evidence; do not generalize from one stratum.
Adapter fidelity	Direct/derived label agreement and downstream differences on the same subset	Annotation conversion cannot change the construct being evaluated.

Table 10: Information required before confirmatory evaluation. Unset values are intentional placeholders, not inferred development defaults.

Item	Required specification
Domain and sources	Versioned taxonomy, decision rule, readiness audit, dataset versions, permissions, input counts, candidate groups: <i>TBD</i>
Supervision	Direct/derived provenance, H_0 – H_3 availability, core dimensions, annotator pool, agreement, time, tie policy: <i>TBD</i>
Partitioning	Original-input/source grouping, deduplication, split sizes, OOD definition, External role, contamination review: <i>TBD</i>
Mining	Source inventory, mappings, dependence groups, anchor size, calibration, strata, frozen policy, human-budget accounting: <i>TBD</i>
Models	Frozen scorer, proposer, tool-model, and generator identities, revisions, decoding, and permissible evidence: <i>TBD</i>
Program and search	Seed, editable components, edit bounds, allocation, development guard, protection criteria, shortlist rule: <i>TBD</i>
Measurement	Repeats $R \geq 3$, precision, completeness, retries, artifact reuse, cache identity, search seeds, stopping rule: <i>TBD</i>
Budgets	Human time, opportunities, model calls/tokens, tools, runtime, deployment cost, training compute, matching rule: <i>TBD</i>
Downstream controls	C1 pool size; C2 fresh generation and refinement; C3 reward, LoRA/GRPO settings, pilot gates and preflight: <i>TBD</i>
Analysis	Primary contrasts, grouped intervals, multiplicity, missingness, reserved metrics, minimum effects, final sample sizes: <i>TBD</i>